DP Biology – A3.1 (HL extension)

Species, keys and barcodes

- What is a species?
- What patterns of genome diversity appear within and between species?

HL statements A3.1.12–15. Textbook pp. 118–124.



Image: a dandelion seed head. Markus Bernet, CC BY-SA 2.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	A3.1.12–13	Explain why the biological species concept fails for some organisms, and why chromosome number is shared within a species
2	A3.1.14–15	Build a dichotomous key for local organisms, and explain how eDNA barcodes identify species

The big picture: naming the hard cases

Clones

No interbreeding: what is a species?

Chromosomes

Why hybrids fail

Tools

Keys and DNA barcodes

The biological species concept relies on **interbreeding**. Some organisms never interbreed, so biologists use other evidence.

Big question

How can a single cup of pond water tell you which fish live there?

Asexual organisms

A3.1.12



Dandelion, Taraxacum officinale. Public domain.

Many dandelions make seeds **without fertilization**: offspring are clones.

Clones never interbreed, so each line could count as a species. Thousands of lines exist; biologists group them by **similarity**.

Others

Bacteria, many fungi, some lizards and aphids.

Bacteria: genes move sideways

A3.1.12



Example

Antibiotic-resistance genes on plasmids spread between different bacterial species.

So

Gene pools are not isolated. Bacteria are grouped by shared features and DNA similarity (rRNA genes).

Chromosome number within a species

A3.1.13

All members of a species share the same **chromosome number**, because homologous chromosomes must **pair in meiosis**.



same number: every chromosome has a partner

odd number: one cannot pair

Key idea

A hybrid of parents with different numbers has unpaired chromosomes; meiosis fails and it is usually infertile.

Chromosome numbers and hybrids

A3.1.13



Human karyotype: 46 chromosomes. NHGRI, public domain.

Species	Chromosomes (2n)
Human	46
Chimpanzee	48
Horse	64
Donkey	62
Mule (hybrid)	63

Try it

Horse gametes have 32 and donkey gametes 31. Why can a mule grow by mitosis but not make gametes by meiosis?

Hybrids show the barrier

A3.1.13



A zebroid: zebra × horse hybrid. Public domain.

Zebras have 32 to 46 chromosomes depending on species; horses 64.

Zebroids grow well but are almost always **sterile**.

Key idea

Sterile hybrids show the parents are separate species.

Practice: hard cases

A3.1.12–13

Explain

Why dandelion clones challenge the species concept.

Explain

Why HGT blurs bacterial species.

Predict

Fertility of a hybrid with 2n = 47.

Use

interbreed, clone, plasmid, homologous, meiosis

Check-in: class 1

A3.1.12–13

1. Explain why the species concept is hard to apply to asexual organisms.
2. Define horizontal gene transfer.
3. Explain why chromosome number is shared within a species.
4. Explain why a mule is infertile.
5. Explain what sterile hybrids show.

How to use this

Answer without your notes, then check against slides 3–9.

Dichotomous keys

A3.1.14

	Feature	Go to
1	Leaf divided into leaflets	2
	Leaf simple	3
2	Leaflets from one point	Horse chestnut
	Leaflets along a stalk	Ash
3	Deep lobes	Oak
	Toothed, not lobed	4
4	Base uneven	Elm
	Base even	Birch

Each step is a **couplet**: two opposite statements.

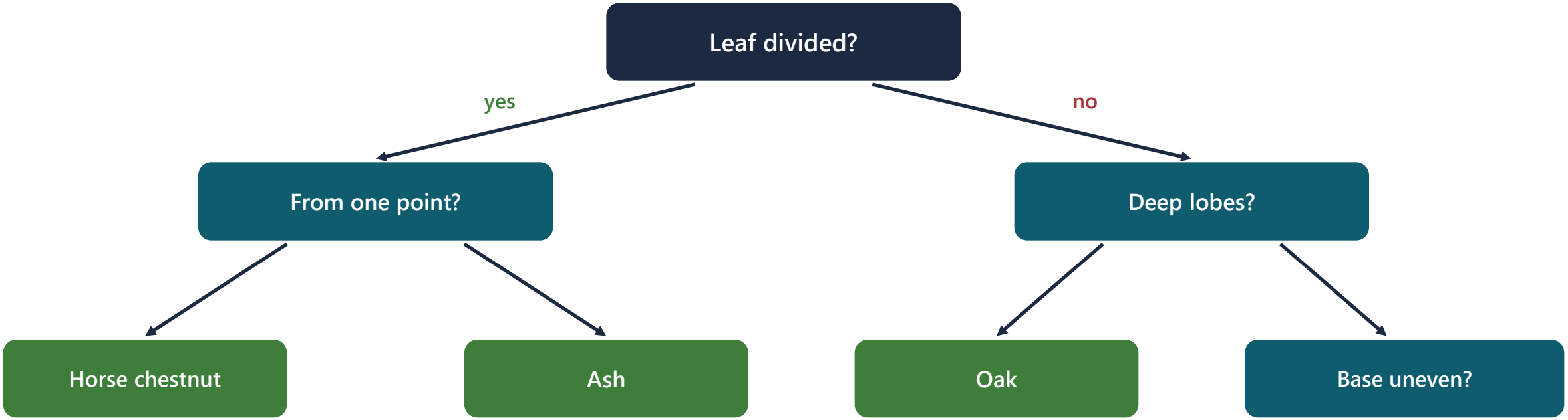
Use clear features that do not vary.

Tips

Avoid size or colour alone; use arrangement, margin, venation.

A key as a branching diagram

A3.1.14

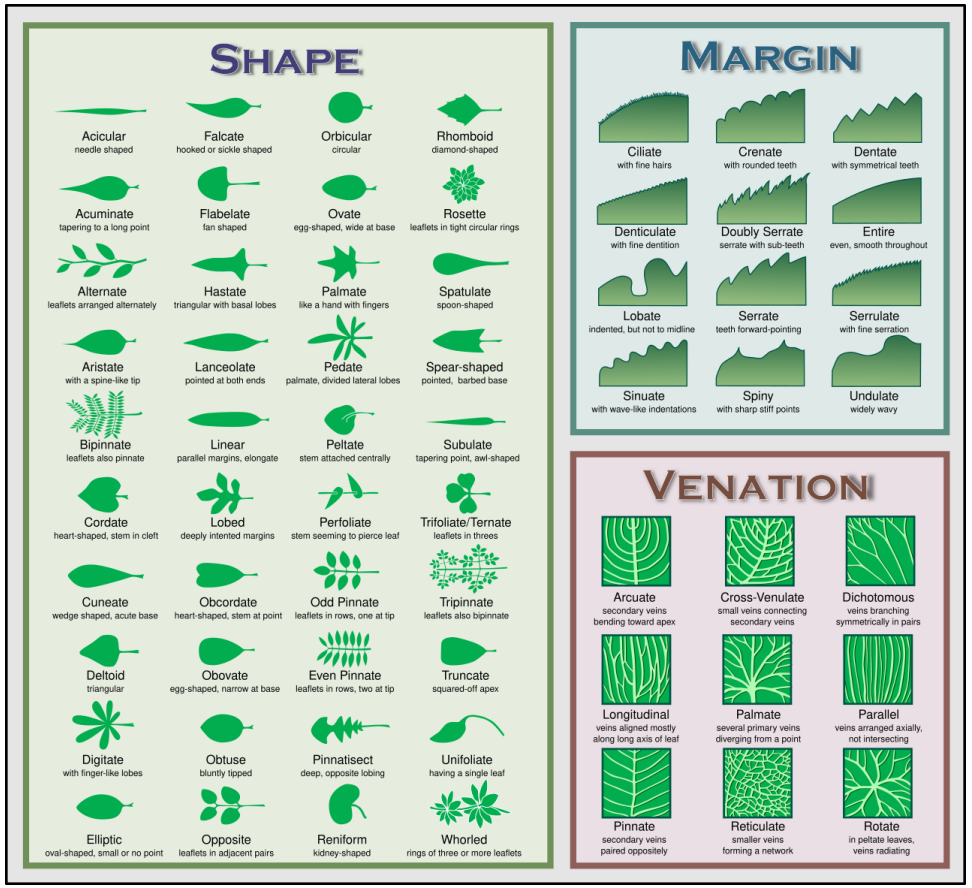


Same key, two forms

A numbered list and a branching diagram hold the same couplets.

Make a key for local species

A3.1.14



Leaf shapes, margins and venation. McSush, CC BY-SA 3.0.

1. Collect leaves or photos from school grounds
2. Identify each with a guide or app
3. Split the group in two with one feature
4. Write couplets until each species is reached
5. Swap keys and test them

Respect

Take one leaf of common plants only; leave animals where they are.

Practice: write the couplets

A3.1.14

Organism	Legs	Wings	Shell
Snail	0	No	Yes
Slug	0	No	No
Beetle	6	Yes	No
Woodlouse	14	No	No

Task

Write a dichotomous key that separates all four, using yes/no or number features.

Environmental DNA and barcodes

A3.1.15



Field sampling in a river. Mukesh Bhendarkar, CC0.

Organisms shed skin, mucus and faeces: **environmental DNA**.

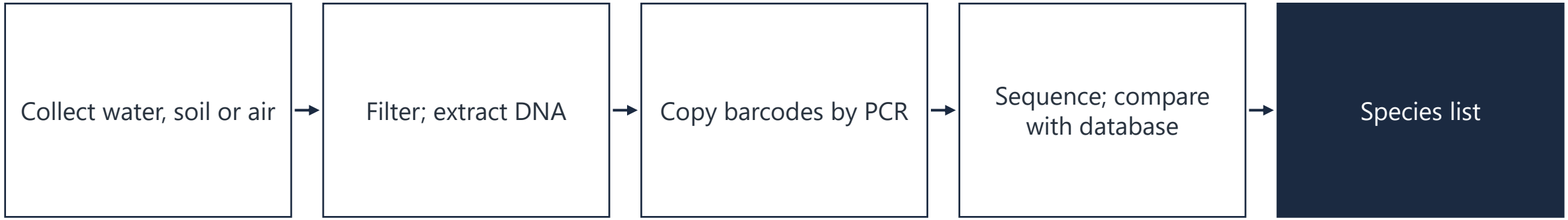
A **barcode** is a short standard DNA section that differs between species (a mitochondrial gene for animals; chloroplast genes for plants).

Big question

One water sample can reveal fish, amphibians and mammals living nearby.

From water sample to species list

A3.1.15



Key idea

Matching barcode sequences against a reference database names the species present, without seeing or catching them.

Why use eDNA?

Strengths

Fast; many species at once; detects rare, hidden or night-active species without catching them; finds invasives early.

Limitations

DNA may drift or persist after the organism leaves; no numbers or health; unknown species cannot be named; contamination risk.

Example

Pond water shows whether great crested newts live there, without disturbing them.

Working with data: eDNA versus netting

A3.1.15

Fish species detected in a lake by two methods. Invented values; pattern matches published comparisons.

Result	Species
Found by both	14
Only by eDNA	6
Only by netting	1

1. Total species and % found by eDNA?
2. Why did eDNA find more?
3. Give one limitation of eDNA.

Practice: keys and barcodes

A3.1.14–15

Explain

Why a key only works for its own organisms.

Outline

The steps of an eDNA survey.

Evaluate

eDNA versus trapping for a rare newt.

Use

couplet, barcode, PCR, reference database

Check-in: class 2

A3.1.14–15

1. Describe a couplet in a key.
2. Give two rules for good keys.
3. Define eDNA.
4. Define a DNA barcode.
5. Give one strength and one limitation of eDNA.

How to use this

Answer without your notes, then check against slides 11–19.

Common mistakes to avoid

Not this	But this
Hybrids are always new species.	Most are infertile: parents are separate species.
A key identifies any organism.	Only those it was written for.
eDNA comes only from dead organisms.	Living organisms shed DNA constantly.

Summary: species, keys and barcodes

A3.1.12–15

	Key idea
A3.1.12	Asexual organisms and HGT challenge the species concept
A3.1.13	Shared chromosome number; sterile hybrids
A3.1.14	Dichotomous keys
A3.1.15	eDNA and barcodes

Check that you can...

1. **Discuss** difficulties with the species concept
2. **Explain** hybrid infertility
3. **Construct** and **use** dichotomous keys
4. **Outline** eDNA and barcoding

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why the biological species concept is difficult to apply to bacteria.
2. State why a mule is usually infertile.
3. Outline how eDNA is used to find which species live in a lake.

Image credits

Dandelion clock.jpg – Markus Bernet – CC BY-SA 2.0

Taraxacum officinale – Köhler–s Medizinal-Pflanzen-135.jpg – Walther Otto Müller – Public domain

NHGRI human male karyotype.png – Courtesy: National Human Genome Research Institute – Public domain

Zebroid.jpg – Olliver at English Wikipedia – Public domain

Leaf morphology no title.svg – derivative work: McSush (talk) Leaf_morphology_no_title.png: – CC BY-SA 3.0

EDNA sampling.jpg – Mukesh Bhendarkar – CC0

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